TIANDONG WANG

SCMS, Fudan University | td_wang@fudan.edu.cn

EDUCATION

Cornell University — Ithaca, NY
Ph.D. in Operations Research — Aug 2014–Aug 2019
Concentration: Applied Probability and Statistics; Minors: Finance and Mathematical Statistics.
M.Sc. in Operations Research — Aug 2014–Jan 2017

Australian National University — Canberra, Australia
Bachelor of Actuarial Studies with Honors in Statistics — Feb 2010–Dec 2013

EMPLOYMENT HISTORY

Prior tenure-track faculty experience: Before joining Fudan University, I completed three full academic years (September 2019–August 2022) as a full-time tenure-track Assistant Professor in the Department of Statistics at Texas A&M University in USA.

Tenure-Track Associate Professor, **Shanghai Center for Mathematical Sciences** — Sep 2025–Present
Fudan University — Shanghai, China

Young Investigator, **Shanghai Center for Mathematical Sciences** — Sep 2022–Sep 2025
Fudan University — Shanghai, China

Tenure-Track Assistant Professor, **Department of Statistics** — Sep 2019–Aug 2022
Texas A&M University — College Station, TX, USA

Full-Time Research Assistant, **Mathematical Sciences Institute** — Dec 2013–Jul 2014
Australian National University — Canberra, ACT, Australia

RESEARCH INTERESTS

Random networks; extreme value theory; network inference; applied probability and stochastic processes; mathematical statistics.

AWARDS AND SCHOLARSHIPS

Second Prize, Shanghai Science and Technology Progress Award — *Shanghai Municipal People's Government, 2025*
Fudan University Outstanding Talent Program (Zhuo Xue) — *Fudan University, 2023*
Shanghai Leading Talent (Overseas) Program (Ling Jun) — *Shanghai Municipality, 2022*
International Statistical Institute (ISI) Elected Member — *International Statistical Institute*, since 06/2022
Runner Up, 2020 Doctoral Dissertation Award for Operations Research in Telecommunications and Network Analytics — *INFORMS Telecommunication & Analytics, 2020*
University Medal in Statistics — *Australian National University, 2013*
Goldman Sachs J B Were Prize — *Australian National University, 2013*
College of Business and Economics Honors Scholarship — *Australian National University, 2012*

GRANTS

1. PI, National Natural Science Foundation of China, General Program, Grant No. 12671644, "Local Topological Inference and Generation Mechanism Identification for High-dimensional Sparse Random Graphs", 2027–2030.
2. PI, National Natural Science Foundation of China, Young Scientists Fund (Type C), Grant No. 12301660, "Heterogeneous Scale-Free Networks and Related Heavy-Tail Phenomena: Modeling and Inference", 2024–2026.

3. Core Team Member, National Key R&D Program of China, Grant No. 2025YFA1016503, Ministry of Science and Technology, 2025–2029.
4. PI, Shanghai Institute for Mathematics and Interdisciplinary Sciences Grant SIMIS-ID-2024-WE, "Multi-level data governance and mechanism analysis of complex education data", 2024–2026.
5. PI, Science and Technology Commission of Shanghai Municipality Grant 23JC1400700, "Multilayer Network Modeling, Analysis and Inference", 2023–2026.
6. PI, NSF DMS-2210685, "Collaborative Research: Models and Inferences for Heterogeneous Interaction Patterns of Social Networks," 2022–2025. Subsequently a Participant after joining Fudan University.
7. Participant, NSFC-7217010183 (PI: Y. Hou, School of Data Science, Fudan University), 2022–2025.
8. Co-PI, T3: Texas A&M Triads for Transformation, Round 4, 2021–2022.

OTHER RESEARCH & PROFESSIONAL ACTIVITIES

1. Supervised a Ph.D. student (Q. Hui) who received funding through the AI for Math and Science Program, Shanghai Institute for Mathematics and Interdisciplinary Sciences (SIMIS), 2026–2028.
2. Entrance Examination Committee, Master's Program in Applied Statistics, Fudan University, 2024–2026 (three consecutive years).
3. Participated in the Ministry of Education's Employment Satisfaction Survey for Double First-Class Graduates, 2024.
4. Participated in the National Data Resources Survey and prepared analytical reports, 2025 and 2026.
5. Selected for an Annual Outstanding Research Project by the Shanghai Municipal Big Data Center; received a letter of appreciation, 2025.

REPRESENTATIVE PUBLICATIONS

Publication Summary. As of August 2026, I have published 1 book and 31 peer-reviewed articles. Since joining Fudan University in 2022, I have published or received acceptance for 14 papers, including two papers in the *Journal of Machine Learning Research* (JMLR 2024; 2026) and recent publications in *Stochastic Processes and their Applications* and *Statistica Sinica*.

1. D. Cirkovic, **T. Wang**. Modeling random networks with heterogeneous reciprocity. *Journal of Machine Learning Research*, 25(10): 1–40, 2024. (Corresponding author)
2. D. Shi, **T. Wang**, Z. Ying. Simultaneous identification of sparse structures and communities in heterogeneous graphical models. *Journal of Machine Learning Research*, 27(148):1–63, 2026. (Corresponding author)
3. D. Cirkovic, **T. Wang** and D.B.H. Cline. Emergence of multivariate extremes in multilayer inhomogeneous random graphs. *Stochastic Processes and their Applications*, 190:104762, 2025. (Corresponding author)
4. D. Cirkovic, **T. Wang** and X. Zhang. Likelihood-based inference for random networks with changepoints. *IEEE Transactions on Network Science and Engineering*, 13: 344–359, 2025. (Corresponding author)
5. **T. Wang** and S.I. Resnick. Degree growth rates and index estimation in a directed preferential attachment model. *Stochastic Processes and their Applications*, 130(2): 878–906, 2020. (First and corresponding author)

PUBLICATIONS

Books

1. **T. Wang**, Y. Sun, Y. Xia, C. Guo, Y. Fu, and X. Huang. *Data Governance and Model Optimization: Methodological Practices Based on Agile Statistics* [in Chinese]. Publishing House of Electronics Industry, 2026. ISBN 978-7-121-53268-9.

Peer-Reviewed Articles

1. D. Shi, **T. Wang**, Z. Ying. Simultaneous identification of sparse structures and communities in heterogeneous graphical models. *Journal of Machine Learning Research*, 27(148):1–63, 2026. (Corresponding author)

2. Q. Hui and **T. Wang**. Extremal dependence networks for mitigating systemic risk via maximum independent sets, *Journal of Complex Networks*, 14(4): cnag005, 2026. (Corresponding author)

3. Z. Han and **T. Wang**. Measuring interlayer dependence of large degrees in multilayer inhomogeneous random graphs, *Physica A: Statistical Mechanics and its Applications*, 682: 131173, 2026. (Corresponding author)

4. Z. Han, **T. Wang** and Z. Ying. Multilayer network regression with eigenvector centrality and community structure. *Statistica Sinica*, 2025, doi: 10.5705/ss.202025.0101. (Corresponding author)

5. D. Cirkovic, **T. Wang** and X. Zhang. Likelihood-based inference for random networks with changepoints. *IEEE Transactions on Network Science and Engineering*, 13: 344-359, 2025. (Corresponding author)

6. **T. Wang** and S.I. Resnick. Distinguishing forms of asymptotic dependence in heavy tailed data. *Statistica Sinica*, 2025, doi: 10.5705/ss.202024.0196. (First and corresponding author)

7. Z. Li, Y. Hou, **T. Wang**. Recent advances on mechanisms of network generation: Community, exchangeability, and scale-free properties. *Wiley Interdisciplinary Reviews: Computational Statistics*, 16(2): e1651, 2024. (Corresponding author)

8. D. Cirkovic, **T. Wang**. Modeling random networks with heterogeneous reciprocity. *Journal of Machine Learning Research*, 25(10): 1–40, 2024. (Corresponding author)

9. **T. Wang** and S.I. Resnick. Random networks with heterogeneous reciprocity levels. *Extremes*, 27(1): 123–161, 2024. (First and corresponding author)

10. D. Cirkovic, **T. Wang** and S.I. Resnick. Preferential attachment with reciprocity: Properties and estimation. *Journal of Complex Networks*, 11(5): cnad031, 2023. (Corresponding author)

11. Y. Yuan, **T. Wang**, P. Zhang and J. Yan. Generating general preferential attachment networks with R package `wdnet`. *Journal of Data Science*, 21(3): 538–556, 2023. (Corresponding author)

12. **T. Wang** and S.I. Resnick. Common growth patterns in regional communication networks: a point process approach. *Journal of Data Science*, 21(3): 446–469, 2023. (First and corresponding author)

13. **T. Wang** and S.I. Resnick. Poisson edge growth and preferential attachment networks. *Methodology and Computing of Applied Probability*, 25(1): 8, 2023. (First and corresponding author)

14. **T. Wang**, J. Yan, Y. Yuan and P. Zhang. Generating directed networks with predetermined assortativity measures. *Statistics and Computing*, 32(5): 91, 2022.

15. J. Wang, Y. Hou, X. Li and **T. Wang**. EVIboost for the estimation of extreme value index under heterogeneous extremes, *Journal of Data Science*, 21(4): 638–657, 2022.

16. **T. Wang** and S.I. Resnick. Measuring reciprocity in a directed preferential attachment network. *Advances in Applied Probability*, 54(3): 718–742, 2022. (First and corresponding author)

17. **T. Wang** and S.I. Resnick. Asymptotic dependence of in- and out-degrees in a preferential attachment model with reciprocity. *Extremes*, 25(3): 417–450, 2022. (First and corresponding author)

18. **T. Wang** and P. Zhang. Hybrid random networks mixing preferential attachment with uniform attachment mechanisms. *Annals of the Institute of Statistical Mathematics*, 74(5): 957–986, 2022. (First and corresponding author)

19. C. Tang, **T. Wang** and P. Zhang. Functional data analysis: An application to COVID-19 data in the United States. *Quantitative Biology*, 10(2):172–187, 2022.

20. B. Das, **T. Wang** and G. Dai. Asymptotic behavior of common connections in sparse random networks. *Methodology and Computing in Applied Probability*, 24: 2071–2092, 2022. (Corresponding author)

21. P. Zhang, **T. Wang** and J. Yan. PageRank centrality and algorithms for weighted, directed networks. *Physica A: Statistical Mechanics and its Applications*, 586: 126438, 2022.

22. P. Zhang, **T. Wang** and S.X. Xie. Meta-analysis of several epidemic characteristics of COVID-19. Journal of Data Science, 18(3): 536–549, 2022.

23. **T. Wang** and J. Yan. Discussion of "On Studying Extreme Values and Systematic Risks with Nonlinear Time Series Models and Tail Dependence Measures". *Statistical Theory and Related Fields*, 5(1): 38–40, 2021.

24. **T. Wang** and S.I. Resnick. Degree growth rates and index estimation in a directed preferential attachment model. *Stochastic Processes and their Applications*, 130(2): 878–906, 2020. (First and corresponding author)

25. H. Drees, A. Janßen, S.I. Resnick and **T. Wang**. On a minimum distance procedure for threshold selection in tail analysis. *SIAM Journal on Mathematics of Data Science*, 2(1): 75–102, 2020. (Authors listed in alphabetical order)

26. P. Wan, **T. Wang**, R.A. Davis, and S.I. Resnick. Are extreme value estimation methods useful for network data? *Extremes*, 23(1): 171–195, 2020. (Corresponding author)

27. **T. Wang** and S.I. Resnick. Consistency of Hill estimators in a linear preferential attachment model. *Extremes*, 22(1): 1–28, 2019. (First and corresponding author)

28. **T. Wang** and S.I. Resnick. Multivariate regular variation of discrete mass functions with applications to preferential attachment networks. *Methodology and Computing in Applied Probability*, 20:1029–1042, 2018. (First and corresponding author)

29. P. Wan, **T. Wang**, R.A. Davis and S.I. Resnick. Fitting the linear preferential attachment model. *Electronic Journal of Statistics*, 11(2):3738–3780, 2017. (Co-first author)

30. **T. Wang** and S.I. Resnick. Asymptotic normality of in- and out-degree counts in a preferential attachment model. *Stochastic Models*, 33(2):229–255, 2017. (First and corresponding author)

31. Y. Fan, P. Griffin, R.A. Maller, A. Szimayer and **T. Wang**. The effects of largest claims and excess of loss reinsurance on a company's ruin time and valuation, *Risks*, 5(1):3, 2017. (Authors listed in alphabetical order)

PREPRINTS

Accepted:

1. Q. Hui, S.I. Resnick, **T. Wang**. Classification of extremal dependence in financial markets via bootstrap inference. *Journal of Time Series Analysis*, 2026. Accepted for publication. (Corresponding author)

Under revision:

1. D. Shi, H. Zhang, **T. Wang**, J. Wang. A multilayer probit network model for community detection with dependent edges and layers. Minor revision submitted to *Journal of the American Statistical Association*, 2026.

2. **T. Wang** and S.I. Resnick. Bootstrap methods for testing asymptotic dependence in multivariate heavy-tailed data. Revision submitted to *Extremes*, 2026.

Submitted:

1. D. Cirkovic and **T. Wang**. Efficient topic model estimation under heavy-tailed document lengths. ArXiv: 2607.24641. Submitted to *Journal of Machine Learning Research*, 2026.

2. Q. Hui and **T. Wang**. Hub neighbor-degree diagnostics for sparse random graphs. ArXiv: 2607.26624. Submitted to *IEEE Transactions on Network Science and Engineering*, 2026.

3. D. Cirkovic, **T. Wang**, D.B.H. Cline. On tail inference in scale-free inhomogeneous random graphs. ArXiv: 2410.22703. Submitted to *Bernoulli*, 2026.

4. F. Fang, S. Bai, **T. Wang**. Structural causal models for extremes: an approach based on exponent measures. ArXiv:2508.00223. Submitted to *Annals of Applied Probability*, 2026.

STUDENTS

Ph.D. Advising — Primary Advisor

1. Daniel Cirkovic, Ph.D., Texas A&M University (chair)
2. Qian Hui, current Ph.D. student, Fudan University (chair)

Ph.D. Dissertation Committees

1. Dapeng Shi (Ph.D. student at Fudan University, committee member)
2. Zhuoye Han (Ph.D. student at Fudan University, committee member)
3. Huiling Liao (Ph.D. student at Texas A&M University, committee member)

Master's Students in Applied Statistics, Fudan University

1. Class of 2024: Yating Kong, Liuxin Song, Xuanqi Tao, Li Zhang, Jiawen Zhou
2. Class of 2025: Wenchao Chen, Chao Ma, Qianqing Tang, Zhuoyang Wu, Jingwen Ye
3. Class of 2026: Hongwei Fan, Zihao Fang, Yinru Long, Xinyu Shen, Jiahao Yang
4. Class of 2027: Dong Fang, Yiyang Geng, Yumo Hu, Ziyi Zhang, Mingjie Zheng

M.S. Students, Texas A&M University

1. Sijing Yu (M.S. student at Texas A&M University, committee member)
2. Vrushabh Bhangod (M.S. student at Texas A&M University, committee member)
3. Andrew Lastinger (M.S. student at Texas A&M University, committee member)
4. Alexander Coulter (M.S. student at Texas A&M University, co-chaired with D.B.H. Cline)

Undergraduate Students in Mathematics, Fudan University

1. Class of 2027: Tianyu Chen, Shuang Li, Yuehuan Lin, Yaoyao Ma, Haoran Zhang, Junji Zhang

TEACHING

SCMS, Fudan University

Instructor, SCMS637003 Foundation of Applied Statistics	Fall 2023, 2024, 2025 & 2026
Instructor, MATH130015 Undergraduate Thesis	Spring 2025
Instructor, MATH130162 Independent Study I	Spring 2025
Instructor, SCMS817002 Complex Networks and Stochastic Analysis	Spring 2023

Texas A&M University

Instructor, STAT 414-501 Mathematical Statistics I	Fall 2021
Instructor, STAT 695-600 Frontier Course on Random Networks	Spring 2021
Instructor, STAT 650-600 Statistical Foundation for Data Science	Fall 2020 & 2021
Instructor, ICPE 689 Data Science Foundation for Energy I (Module 1)	Fall 2020
Instructor, STAT 611-600 Theory of Inference	Spring 2020, 2021 & 2022
Instructor, STAT 689-600 Probability and Statistics for Data Science	Fall 2019

Cornell University

Teaching Assistant, ORIE 6500 Applied Stochastic Processes	Spring 2019
Teaching Assistant, ENGRD 2700 Basic Engineering Probability and Statistics	Fall 2018
Teaching Assistant, ORIE 5640 Statistics for Financial Engineering	Spring 2018
Teaching Assistant, ENGRD 2700 Basic Engineering Probability and Statistics	Fall 2014

Australian National University

Academic Tutor, STAT 2008/6038 Regression Modeling	Spring 2012

CONFERENCE ORGANIZATION

Organizer, *One World Extremes Seminar*, 2025–2027, Global Online Events.

Session organizer, *Extreme Value Analysis 2025*, Jun 2025, Chapel Hill, USA.

Chair of organizing committee, *International Conference on Extremes, Statistics, and Quantitative Risk Management*, July 29–31, 2024, Shanghai Center for Mathematical Sciences, Fudan University.

Chair of organizing committee, *Workshop on Statistics and Machine Learning*, August 12–14, 2023, Shanghai Center for Mathematical Sciences, Fudan University.

Chair of organizing committee, *SETCASA Poster Competition*, May 2022, Department of Statistics, Texas A&M University.

Session organizer, *ICSA Applied Statistics Symposium*, Dec 2020, Houston, USA.

INVITED TALKS

"Emergence of Multivariate Extremes in Multilayer Inhomogeneous Random Graphs"
Stochastic Processes and their Applications (SPA 2026), June 2026, Ithaca, USA.
"Emergence of Multivariate Extremes in Multilayer Inhomogeneous Random Graphs"
Joint Meeting of the 13th Days of Econometrics for Finance and the Workshop on Recent Advances in Extreme Value Analysis (JEF/WEVA 2026), April 2026, Rabat, Morocco.
"Bootstrap Methods for Testing Asymptotic Dependence in Multivariate Heavy-Tailed Data"
Joint Fudan-Vienna Workshop on Applied Mathematics and Data Science, June 2025, Vienna, Austria.
"Distinguishing Forms of Asymptotic Dependence in Heavy Tailed Data"
Mathematics, Statistics, and Geometry of Extreme Events in High Dimensions, August 2024, Oberwolfach, Germany.
"Testing for Strong VS Full Dependence"
Fifth International Workshop on Statistical Analyses of Multi-Outcome Data, July 2024, Salzburg, Austria.
"Testing for Strong VS Full Dependence"
Joint Fudan-Vienna Workshop on Applied Mathematics and Data Science, February 2024, Shanghai, China.
"Changepoint Detection in Preferential Attachment Networks"
International Forum of Statistics 2023, July 2023, Beijing, China.
"Changepoint Detection in Preferential Attachment Networks"
ICSA International Conference, July 2023, Hong Kong, China.
"Random Networks with Heterogeneous Reciprocity"
INFORMS Applied Probability Conference, June 2023, Nancy, France.
"Random Networks with Heterogeneous Reciprocity"
13th Extreme Value Analysis Conference, June 2023, Milan, Italy.
"Asymptotic Dependence of In- and Out-Degrees in a Preferential Attachment Model with Reciprocity"
New England Statistics Symposium, May 2022, University of Connecticut, Connecticut, USA.
"Reciprocity and Large Degree Dependence in a Preferential Attachment Network"
12th Extreme Value Analysis Conference, Online, June 2021.
"A Linear Preferential Attachment Model with Poisson Growth"
ICSA Applied Statistics Symposium, Dec 2020, Houston, USA.
"A Directed Linear Preferential Attachment Model with Poisson Measurement"
Statistics Colloquium, University of Connecticut, Nov 2020.
"Heavy Tail Phenomena in Preferential Attachment Networks"
INFORMS Telecommunication & Network Analytics, Oct 2020.
"A Directed Linear Preferential Attachment Model with Poisson Measurement"
Statistics Colloquium, Fudan University, June 2020.
"Degree Growth and Index Estimation in a Preferential Attachment Network"
11th Extreme Value Analysis Conference, July 2019, Zagreb, Croatia.
"Estimation in Preferential Attachment Networks"
Shanghai Center of Mathematical Sciences, Fudan University, June 2019, Shanghai, China.
"Estimation in Preferential Attachment Networks"

13th INFORMS Data Mining and Decision Analytics Workshop, November 2018, Phoenix, USA.
"Are Extreme Value Estimation Methods Useful for Network Data?"
Joint Statistics Meeting, July 2018, Vancouver, Canada.
"Consistency of Hill Estimators in a Linear Preferential Attachment Model".
Self-Similarity, Long-Range Dependence and Extreme, June 2018, BIRS-CMO, Oaxaca, Mexico.
"Multivariate Regular Variation of In- and Out-Degrees in Preferential Attachment Networks".
Workshop on Lévy processes and time series: in honor of Peter Brockwell and Ross Maller, September 2017, Ulm University, Germany.
"Multivariate Regular Variation of In- and Out-Degrees in Preferential Attachment Networks".
10th Extreme Value Analysis Conference, June 2017, TU Delft, Netherlands.
"Asymptotic Normality of In- and Out-Degree Counts in a Preferential Attachment Model".
New England Statistics Symposium, April 2017, University of Connecticut, Storrs, Connecticut, USA.

EDITORIAL AND ACADEMIC SERVICE

- Scientific Committee Member, Extreme Value Analysis 2025.
- Board Member, One World Extremes Seminar, 07/2025 to 06/2027.
- President, Southeastern Texas Chapter of ASA (SETCASA), 02/2022 to 01/2023.
- Junior Liaison, National Institute of Statistical Sciences (NISS), 01/2021 to 08/2022.
- Colloquium Chair, Department of Statistics, Texas A&M University, 01/2021 to 12/2021.
- Vice President, Southeastern Texas Chapter of ASA (SETCASA), 02/2021 to 01/2022.
- Secretary, Southeastern Texas Chapter of ASA (SETCASA), 02/2020 to 01/2021.
- Referee for: Annals of Statistics, Journal of the American Statistical Association, Journal of the Royal Statistical Society Series B, IEEE Transactions on Network Science and Engineering, IEEE Transactions on Pattern Analysis and Machine Intelligence, Statistical Science, Extremes, Journal of Business and Economic Statistics, Insurance: Mathematics and Economics, Annals of Operations Research, Journal of Applied Probability, Stat, Electronic Communications in Probability, Journal of Time Series Analysis, Mathematical Reviews.

Updated on September 9, 2026